

A Fine-tuned CLIP-RN50 visual encoder

shape notation: $[B \times C \times H \times W]$

ANTI-ALIASED BOTTLENECK

1×1 Conv $\rightarrow$ BN $\rightarrow$ ReLU $\rightarrow 3 \times 3$ Conv $\rightarrow$ BN $\rightarrow$ ReLU
 $\rightarrow$ optional AvgPool $\rightarrow 1 \times 1$ Conv $\rightarrow$ BN
 $\rightarrow$ Residual Add $\rightarrow$ ReLU

SPATIAL ATTENTION POOL2D

49 spatial tokens + positional embedding
 Q/K/V Linear $2048 \rightarrow 2048 \rightarrow 32$ -head self-attention
 $\rightarrow$ projection $2048 \rightarrow 1024$
 parallel residual 1×1 Conv $2048 \rightarrow 1024 + \text{BN} \rightarrow \text{Add} + \text{ReLU}$
 $V_{\text{raw}} [B \times 1024 \times 7 \times 7] \rightarrow 1 \times 1$ Conv $1024 \rightarrow 512 + \text{BN} + \text{ReLU}$
 $V_5 [B \times 512 \times 7 \times 7]$

Layer 4: Bottleneck $\times 3 \rightarrow [B \times 2048 \times 7 \times 7]$

Layer 3: Bottleneck $\times 6 \rightarrow V_4 [B \times 1024 \times 14 \times 14]$

Layer 2: Bottleneck $\times 4 \rightarrow V_3 [B \times 512 \times 28 \times 28]$

Layer 1: Bottleneck $\times 3 \rightarrow [B \times 256 \times 56 \times 56]$

Stem: $3 \times 3/s_2 \rightarrow 32; 3 \times 3 \rightarrow 32; 3 \times 3 \rightarrow 32; 3 \times 3 \rightarrow 64; \text{AvgPool}/s_2$
 $[B \times 64 \times 56 \times 56]$

$x' [B \times 3 \times 224 \times 224]$

B Multi-scale alignment and CARAFE

all aligned features: $[B \times 512 \times 14 \times 14]$

$V_3 [B \times 512 \times 28 \times 28]$

AvgPool $2 \times 2/s_2$

F_3

$V_4 [B \times 1024 \times 14 \times 14]$

$3 \times 3: 1024 \rightarrow 512 + \text{BN/ReLU}$
 $3 \times 3: 512 \rightarrow 512 + \text{BN/ReLU}$

F_4

$V_5 [B \times 512 \times 7 \times 7]$

CARAFE $\times 2$

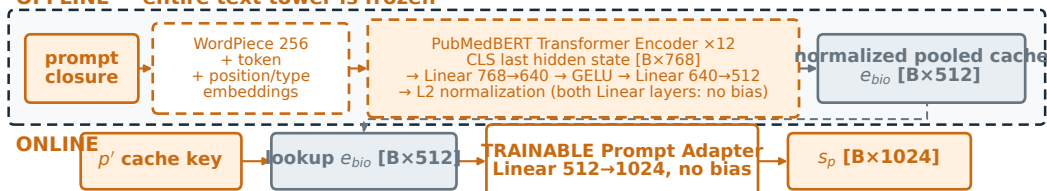
F_5

CARAFE CONTENT-AWARE REASSEMBLY

1×1 Conv $512 \rightarrow 128 \rightarrow 3 \times 3$ Conv $128 \rightarrow 36 \rightarrow \text{PixelShuffle} \times 2 \rightarrow \text{nine } 3 \times 3 \text{ reassembly weights} \rightarrow \text{Softmax}$
 $\rightarrow \text{content-aware reassembly} \rightarrow 1 \times 1$ Conv $512 \rightarrow 512$

C Frozen pooled BioMedCLIP text path

OFFLINE — entire text tower is frozen



ONE TRANSFORMER ENCODER ($\times 12$)

12-head self-attention: Q/K/V Linear $768 \rightarrow 768 \rightarrow \text{attention output Linear } 768 \rightarrow 768 \rightarrow \text{Add} \& \text{LayerNorm}$
 $\rightarrow \text{Linear } 768 \rightarrow 3072 \rightarrow \text{GELU} \rightarrow \text{Linear } 3072 \rightarrow 768 \rightarrow \text{Add} \& \text{LayerNorm}$